DS5020 Mini Project Memo

Product recommendations and customer groups

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For: Q4 planning meeting

Summary

Both things you asked for are ready. The new “also bought” suggestions find a product the customer really bought next **40% of the time, against 27% for a best-seller list**. The whole nightly job runs in **3.5 seconds** and stores about **4 MB**. For marketing, our customers fall into **six groups** defined by what they like to buy.

1. What I did and which method I chose

The old prototype compared every customer with every product: a table of 5,473 customers × 3,687 products, 161 MB when stored in full, even though 98% of it is empty because most customers buy only a few products. I used a method called **Truncated SVD**. It boils that table down to 50 “buying patterns”, such as “buys candle holders and lanterns” or “buys Regency tea sets”, and describes every customer and every product by those patterns. I also tested the alternative, **PCA**. It recommended equally well (40% vs 40%) but took three times as long and needs the full-size table again when making suggestions, so I kept SVD.

2. Is it better than showing our best sellers?

Yes. I hid the last two months of orders (9 Oct to 9 Dec 2011), built the model on the earlier data only, and checked whether each customer’s top 10 suggestions included a product they actually went on to buy for the first time. I tested this on 1,913 returning customers:

Approach	Customers with a correct suggestion in their top 10
My model	**40%**
Top 10 best sellers (the simple alternative)	27%

The suggestions also make sense to a person: a white heart T-light holder suggests other T-light holders and lanterns, and a Regency cake stand suggests matching Regency teacups and a teapot. The model learned this from orders alone, without reading product names.

3. How big and how fast

- **Size:** 161 MB for the full table, down to 3.7 MB for the 50 patterns plus 0.4 MB of ready-made top-10 lists.
- **Speed:** the whole nightly job (learning the patterns, building “also bought” lists for all 3,687 products and personal lists for all 5,473 customers) takes 3.5 seconds on a free cloud machine.

4. What we lose by compressing

The 50 patterns keep the main buying habits but drop individual one-off orders; in numbers, they keep about a third of the detail in the data. That loss does not hurt the suggestions, which beat the best sellers anyway. I also leave out products bought by fewer than 10 customers: they account for about 1 in 10 next purchases, and the model will not suggest them.

5. Customer groups for marketing

- **Vintage Tea Party** (8%)
- **Kids’ Crafts & Games** (19%)
- **Cosy Comfort Gifts** (23%, smallest orders)
- **Party Supplies** (14%, most customers outside the UK)
- **Bags & Storage Bulk Buyers** (15%, largest orders)
- **Romantic Home Décor** (22%).

The groups overlap: many shops buy across several themes, so read each group as the customer’s main interest. The country separates them very little, because 81-98% of each group is UK-based. The map and summary table contain no customer IDs.

6. What could go wrong

- **New customers** (16% of recent buyers) have no history, so they would see best sellers until they order.
- **Seasonality:** I tested on October-December, so results may be weaker in quieter months.
- **Repetitive suggestions:** lists often show variants of the same item (other cake cases), which may feel repetitive on a product page.
- **Settings chosen on the same test months:** the result may be slightly optimistic.

7. Recommended next steps

1. Run a live A/B test of the box against best sellers on a share of product pages.
2. Limit lists to two items per product family.
3. Retrain monthly, and confirm the results on a separate set of months.
4. Show best sellers to new customers until they place a first order.